A PLANE CURVE OF DEGREE D HAS TRIVIAL GLOBAL FUNCTIONS AND ARITHMETIC GENUS (D - 1)(D - 2)/2

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the graded sequence, the sheafified sequence, the cohomology transfer to the plane, and both long-exact identifications with the dimension count are all retained verbatim.

Editorial note. The label lemma-genus-plane-curve names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The plane-curve setup. Same chapter, Lemma lemma-equation-plane-curve. Fixes the closed subscheme Z of the projective plane with principal ideal generated by the homogeneous form F of degree d.
- Graded-to-sheaf exactness and projective-space cohomology. Constructions, Lemma constructions-lemma-proj-sheaves; Cohomology of Schemes, Lemmas coherent-lemma-cohomology-projective-space-over-ring and coherent-lemma-relative-affine-cohomology. Sheafifies 0 -> S(-d) -> S -> S/(F) -> 0 to 0 -> O(-d) -> O -> i_*O_Z -> 0, supplies the cohomology of O and O(-d) on the plane, and moves cohomology from Z to the ambient projective space.

Source reference. Original lemma 01XT: sources/stacks-chow/tags.txt, source lines 2532–2532. Prerequisite pointer invoked at proof line(s) 1929; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 089W: sources/stacks-chow/tags.txt, source lines 10511–10511. Prerequisite pointer invoked at proof line(s) 1932; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01M7: sources/stacks-chow/tags.txt, source lines 2112–2112. Prerequisite pointer invoked at proof line(s) 1919; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0BYB: sources/stacks-curves/pinned-original-source.tex, source lines 1859–1859. Prerequisite pointer invoked at proof line(s) 1905; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $Z \subset \mathbf{P}^2_k$ be as in Lemma 0BYB and let $I(Z) = (F)$ for some $F \in k[T_0, T_1, T_2]$. Then $H^0(Z, \mathcal{O}_Z) = k$ and the genus of Z is $(d-1)(d-2)/2$ where $d = \deg(F)$.*

Proof. Let $S = k[T_0, T_1, T_2]$. There is an exact sequence of graded modules

$$0 \to S(-d) \xrightarrow{F} S \to S/(F) \to 0$$

Denote $i : Z \to \mathbf{P}^2_k$ the given closed immersion. Applying the exact functor $\widetilde{}$ (Constructions, Lemma 01M7) we obtain

$$0 \to \mathcal{O}_{\mathbf{P}^2_k}(-d) \to \mathcal{O}_{\mathbf{P}^2_k} \to i_*\mathcal{O}_Z \to 0$$

because F generates the ideal of Z. Note that the cohomology groups of $\mathcal{O}_{\mathbf{P}^2_k}(-d)$ and $\mathcal{O}_{\mathbf{P}^2_k}$ are given in Cohomology of Schemes, Lemma 01XT. On the other hand, we have $H^q(Z, \mathcal{O}_Z) = H^q(\mathbf{P}^2_k, i_*\mathcal{O}_Z)$ by Cohomology of Schemes, Lemma 089W. Applying the long exact cohomology sequence we first obtain that

$$k = H^0(\mathbf{P}^2_k, \mathcal{O}_{\mathbf{P}^2_k}) \longrightarrow H^0(Z, \mathcal{O}_Z)$$

is an isomorphism and next that the boundary map

$$H^1(Z, \mathcal{O}_Z) \longrightarrow H^2(\mathbf{P}^2_k, \mathcal{O}_{\mathbf{P}^2_k}(-d)) \cong k[T_0, T_1, T_2]_{d-3}$$

is an isomorphism. Since it is easy to see that the dimension of this is $(d-1)(d-2)/2$ the proof is finished. □